Review for Exam 5

1. Study your notes. Go over the material. Cover up the main words and try to figure them out from the definitions. Be sure you can spell them and know what they mean. There is a word list at the end of each chapter. Do the homework. The assignments for these two chapters is not particularly long.

Chapter 20 Enzymes

1. An enzyme is a protein; it is a biological catalyst. It works by speeding up a reaction, sometimes by phenomenal rates. It lowers the energy of activation. It finds an alternative pathway for the reaction. It does *not* alter the net energy of the reaction (ΔE or ΔH). It participates in the reaction, but is not permanently consumed by it.

2. What is the difference between absolute and relative specificity?

3. Terms: enzyme, substrate, regulation, co-factor, co-enzyme, apoenzyme, turn-over rate, zymogen, isozymes (=isoenzymes), allosteric site, inhibitor, modulator.

4. Six classes of enzyme functions. Know what each class does.

5. How are enzymes named?

6. General enzyme mechanism: two models for enzyme mechanism? How similar? different?

7. What factors affect enzyme activity? How does a plot of activity versus "factor" look for each factor?

8. What is enzyme inhibition? irreversible? reversible? Generally, how do competitive and noncompetitive inhibition work? How do inhibitors compare with respect to (1) substrate concentration, (2) position of attachment, and (3) resemblance of inhibitor to substrate

9. How does enzyme regulation work? What is feedback inhibition?

Chapter 21 Nucleic acids and Protein Synthesis

1. What are nucleic acids? what is a nucleotide? Given the structures of the sugar, the phosphate and the five bases, be able to draw any polynucleotide sequence, be able to draw a "cartoon."
2. Order is 5′ to 3′ for phosphodiester linkage. Remember that for writing sequences.
3. Watson and Crick, double helix, base pairing, hydrogen bonding, complementary strands
4. DNA replication
5. List the differences between DNA and RNA.
6. What are the three types of RNA, their abbreviations, and what do they do? What is the ultimate function of DNA?
7. What is the "genetic code"? what is a gene? what is a codon?
8. What are introns and exons?
9. What is the Central Dogma of Molecular Biology? What are replication, transcription, and translation? How (what steps) does it all take place? Be able to follow this to a polypeptide.
10. What are the three kinds of DNA mutations? What is the reading frame?

Chapter 21 – Nucleic Acids and Protein Synthesis

<u>nucleic acid</u> – biological molecules which serve to pass genetic information from generation to generation

1. RNA - ribonucleic acid 2. DNA - deoxyribonucleic acid
both are polymers made up of *nucleotides*.
Nucleotides are made up of:

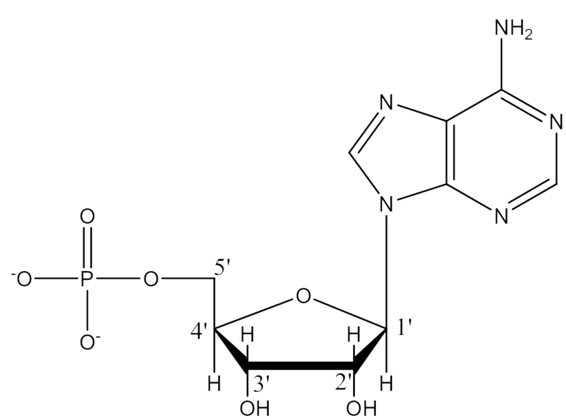
1. nitrogenous bases
   pyrimidines - N-heterocycle with one ring
      uracil (U) - found only in RNA
      thymine (T) - found only in DNA
      cytosine (C) - found in both
   purines - N-heterocycle with two rings
      adenine (A) - found in both
      guanine (G) - found in both

<u>Hydrogen Bonding</u>
   A:U (in RNA)
   A:T (in DNA)
   C:G

2. sugar - either ribose (RNA) or deoxyribose (DNA) with an -OH group at 2' position replaced with an -H (see text p. 671)

3. phosphate group - usually deprotonated

These three parts combine by splitting out 2 water molecules, one between phosphate and sugar and the other between sugar and base. The example below uses the base adenine (in the nucleotide form, now called adenosine monophosphate = AMP)



Nucleotides link together to form the DNA (or RNA) polymer
1. The backbone consists of alternating sugar-phosphate-sugar-phosphate-sugar-phosphate groups formed with a 5'→3' *phosphodiester* linkage (see p. 673).

2. The bases are only attached to the sugars, and at the 1' position.
For DNA, a *double helix* is formed from two strands:

1. Two individual strands are held together with hydrogen bonds between A:T and C:G.
2. The two strands wind around each other to form a double helix. (See p. 674).
3. A and T, and C and G are called *base pairs*, or *complementary base pairs.*
4. The two strands run in opposite directions, that is, they are *antiparallel*. One runs in the 5'→3' direction, and the other in the 3'→5' direction.
5. When DNA sequences are listed, *they are always written (in English) from left to right from the 5'→3' direction.*
6. Once one strand's sequence of bases is known, the other strand is specifically established, since they must be *complements*, relying on the formation of the base pairs.

DNA can do two things: it can be replicated (exact copies made of itself when cells divide), or it can direct the synthesis of proteins. **Replication** and **transcription**.

DNA Replication

When cells divide, the DNA in the chromosomes in the nuclei must be copied so that each daughter cell has a complete set. (See Fig 21.9 and Fig 21.10, p. 676-677).
1. The two strands of DNA unwind at a given point: the *replication fork*.
2. Both strands get their complement bases attached, but working in opposite directions (p. 678).
3. The enzyme *DNA polymerase* binds the nucleotides together to create two new strands, one for each half of the original, unwound DNA.
4. Other enzymes finish joining and checking the new strands.
5. The DNA helices re-wind.
6. The two complete DNA double helices are *each* made from one original strand and one created strand; this is referred to as *semiconservative replication*. (semi=half, conserved from the original)

DNA polymerase is an enzyme for a natural process; it is also used in the laboratory in the process called PCR (polymerase chain reaction), which is used to amplify (make many identical copies) of a DNA sample. This is used in (1) research in genetic diseases, (2) detection of bacterial or viral contamination or disease, (3) forensic science, and (4) cloning of extinct, historical, or archeological samples.

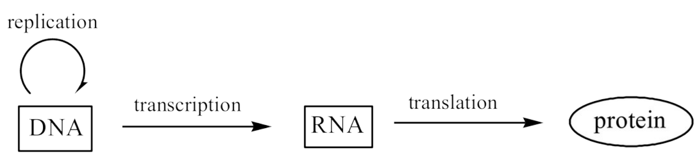
RNA differs from DNA in that (generally)
1. RNA uses ribose where DNA uses deoxyribose.
2. RNA uses uracil where DNA uses thymine.
3. RNA is principally single-stranded (though it does have some loops); DNA is double-stranded.
4. RNA is found in the nucleus, cytoplasm, mitochondria; DNA is only found in the nucleus.
5. RNA is relatively transitory; DNA is relatively permanent.
6. The purpose of RNA is to synthesize proteins; the purpose of DNA is to hold safe the instructions for the synthesis of proteins.

There are three types of RNA:
1. messenger RNA: mRNA - the temporary copy of the genetic material in the DNA; copies DNA in nucleus, goes to cytoplasm where protein synthesis takes place.
2. ribosomal RNA: rRNA - makes up most of the structure of organelles in the cytoplasm called *ribosomes*; this is the "factory" where protein synthesis takes place.

3. <u>transfer RNA</u>: <u>tRNA</u> - small RNA molecules which pick up one amino acid and carry it to the ribosome where the protein is assembled; there is a different tRNA for each of the 20 common amino acids.

The Genetic Code

The sequence of bases on DNA "spells out" (codes for) the sequence of amino acids that make up a protein. A section of a chromosome that is the complete instruction for a protein is called a *gene*. That section of DNA gets copied (in complementary form) onto mRNA, which travels to the ribosome, where tRNAs bring the correct amino acids in the correct order to build a protein. Bases are read in *three base* groups, called a *base triplet* or a *codon*, each one calls for an amino acid (see genetic code, p. 687). The Central Dogma of Molecular Biology is that <u>information flows in one direction, from DNA to RNA to proteins</u>.



The genetic code is (almost) always the same for every living thing. There is code degeneracy, a kind of redundancy wherein there can be more than one codon for each amino acid. This helps in case of reading errors. There are also STOP codes for the termination of protein synthesis.

NOTE: copying the genetic information from DNA to RNA is called *transcription*; but using that information in RNA to make proteins requires a *change of language* (from nucleic acids into amino acids) and so it is called *translation*.

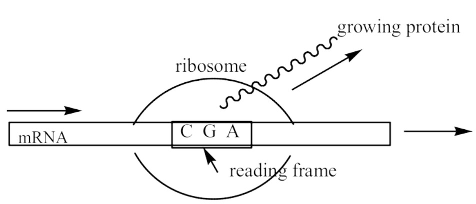
A protein is needed. So,

1. DNA unwinds in the area around the gene that codes for the protein. One strand is used.
2. A mRNA moves into position next to DNA and "copies" it by forming itself into the complement of the DNA strand with the appropriate base pairing. *Note they are antiparallel*.

suppose DNA is 5' G C A A C T T G 3'
then the RNA is 3' C G U U G A A C 5'

which must be correctly written 5'→3' so the RNA sequence is C A A G U U G C

3. The original DNA rewinds and is restored to normal shape in the nucleus. mRNA moves to the ribosome in the cytoplasm. The ribosome is made up mostly of rRNA; it is a "factory" for protein synthesis.
4. The mRNA feeds like a tape through the ribosome. (See Fig 21.20, p. 689)



5. As each codon (3 bases in a row shown above as C G A) passes through the reading frame, an amino acid is called for.
6. A tRNA picks up the correct amino acid and delivers it to the ribosome as the protein gets longer. One end of the tRNA holds the amino acid, the other end has an *anticodon* site, which matches up to the base sequence on mRNA. (See p. 682 for structure of tRNA).
7. As the tRNAs continue to deliver amino acids, the polypeptide grows into a protein.
8. A STOP code ends the process, and the protein is clipped off and released.
9. The mRNA degrades quickly.

HIV is a retrovirus; its genetic material is made of RNA (not DNA). It causes infected cells to re-write its own DNA. This is a violation of Central Dogma, since the information flows backwards from RNA to DNA (hence *retro*virus).

Sometimes one of the bases in the codon may be mis-read; if so, it is best if the error occurs in the third base, may still get the correct amino acid. Note, for example, that (p. 687) the base sequence CAC and CAU both still code for histidine. There are also proofreading enzymes that help to correct errors. However, reading errors cannot overcome the problem of *mutation* - a permanent change in the DNA. A mutation can be very bad, or it can be relatively harmless.

1. insertion - an extra base is inserted in the middle of a proper sequence
2. deletion - a base is deleted from the middle of a proper sequence
3. substitution - a given base is substituted for with a different base

Insertions and deletions disrupt the reading frame since all genetic "words" are three letters, but substitutions may be much less disastrous.
Consider the all three-letter sentence below:

T H E F A T C A T R A N F O R T H E R A T

Read in three-letter groups, it makes sense. But insert a Q as the fourth letter, and the remainder of the words make no sense at all:

T H E Q F A T C A T R A N F O R T H E R A T

But substitute the ninth letter with an R, and while the sense is corrupted, some of it still can be interpreted:

T H E F A T C A R R A N F O R T H E R A T

Practice: transcription and translation
Given a DNA sequence of 5' GCCATATTG 3'
What is the mRNA sequence which would be formed? 3' 5'

How would this RNA sequence be properly written? ______________________________
What is the polypeptide sequence which would be assembled? ______________________________